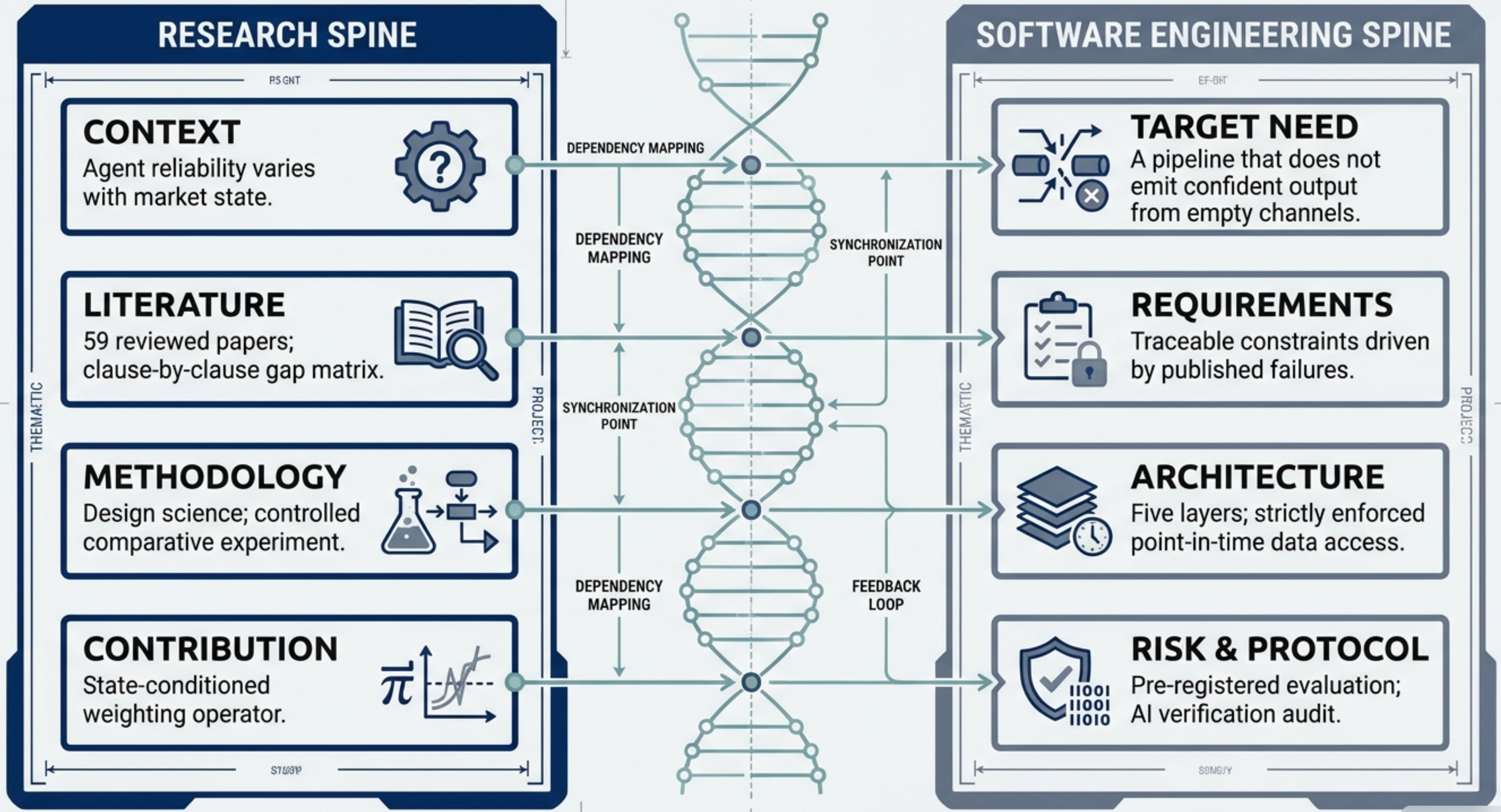
Market-State-Conditioned Orchestration of Heterogeneous Agents for Financial Decision Support

A Colombo Stock Exchange Study

This project investigates how the influence of methodologically heterogeneous financial agents should be set when the reliability of their evidence channels varies with market conditions...
channels varies with market conditions...

Shaahid Fathul | Supervisor: Guhanathan Poravi | 6SENG010C Project Proposal

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RESEARCH SPINE
CONTEXT
Agent reliability varies with market state.
LITERATURE
59 reviewed papers; clause-by-clause gap matrix.
METHODOLOGY
Design science; controlled comparative experiment.
CONTRIBUTION
State-conditioned weighting operator.
DEPENDENCY MAPPING
DEPENDENCY MAPPING
SYNCHRONIZATION POINT
SYNCHRONIZATION POINT
DEPENDENCY MAPPING
FEEDBACK LOOP
SOFTWARE ENGINEERING SPINE
TARGET NEED
A pipeline that does not emit confident output from empty channels.
REQUIREMENTS
Traceable constraints driven by published failures.
ARCHITECTURE
Five layers; strictly enforced point-in-time data access.
RISK & PROTOCOL
Pre-registered evaluation; AI verification audit.

The Problem: When Agents Fail Silently

Pavithya et al. (2021)

Two heterogeneous models on the CSE. Combination was fixed once at a 3:1 ratio based on a different task. Realized influence drifted randomly based on article count.

AlphaAgents

An entire agent was deleted by hand because its input channel was too thin across 15 megacaps. A human noticed and intervened; the system had no way to.

FinAgent

Tuned to large-cap dynamics that then changed. Auxiliary strategies returned 16% **against buy-and-hold's 29%, with nothing in the design able to notice the regime shift.**

The failure mode is worse than a weak signal: agents do not abstain. A thin channel produces confident output about nothing.

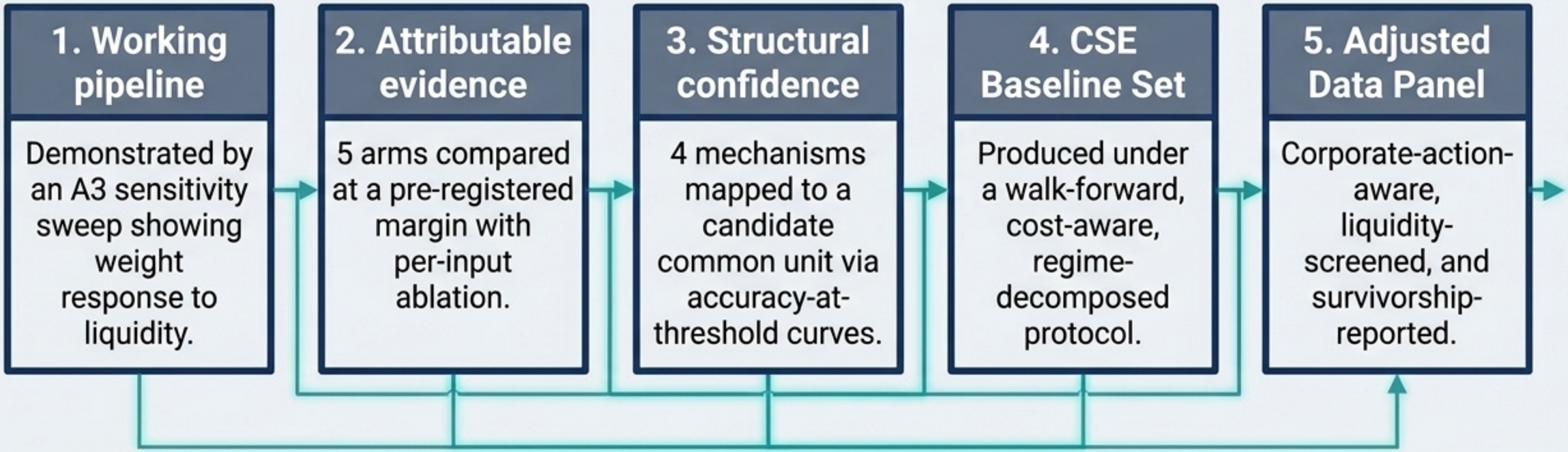
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The Research Gap: Clause-by-Clause Contrast

	Continuous Weighting	Varies Influence (not selection)	Methodologically Heterogeneous	Single Shared Decision	Measured Exogenous Market State
HedgeAgents	✓	✓	✗ Weights assets, not methods.	✓	✓
MacroHFT	✓	✓	✗ Architecturally identical networks (no heterogeneity).		✓
TRIAG	✓	✓	✓	✓	✗ State vector has no environment block.
FFORMA	✓	✓	✓	✓	✗ Conditions on properties of the series being forecast, not exogenous market state.
The Proposal	✓	✓	✓	✓	✓

Synthesis: The project is a defensible generalisation of FFORMA, adapted to market state rather than time-series properties.

Goal: **Design, implement, and evaluate a market-state-conditioned orchestration operator for multi-agent systems, establishing under what conditions it earns its place against fixed and performance-reactive baselines.**



The Mechanism: The Orchestration Operator

Measured exogenous state: Liquidity, Volatility level, and Volatility Persistence.

$$w_{i,t} = f(L_t, V_t, P_t, C_{i,t})$$

Internal state: Structurally computed confidence (not self-reported).

Market state (liquidity, volatility, persistence measured from the market, not from any agent)

- Technical
- Fundamental
- Volatility
- Liquidity

Weighting operator

$$w_{i,t} = f(L_t, V_t, P_t, C_{i,t})$$

1. Listen more to the channel that is working...
2. Abstain when a channel is empty...
3. Remove what cannot be read...

deliberately low capacity: a small, readable function, not a network

One weight per agent

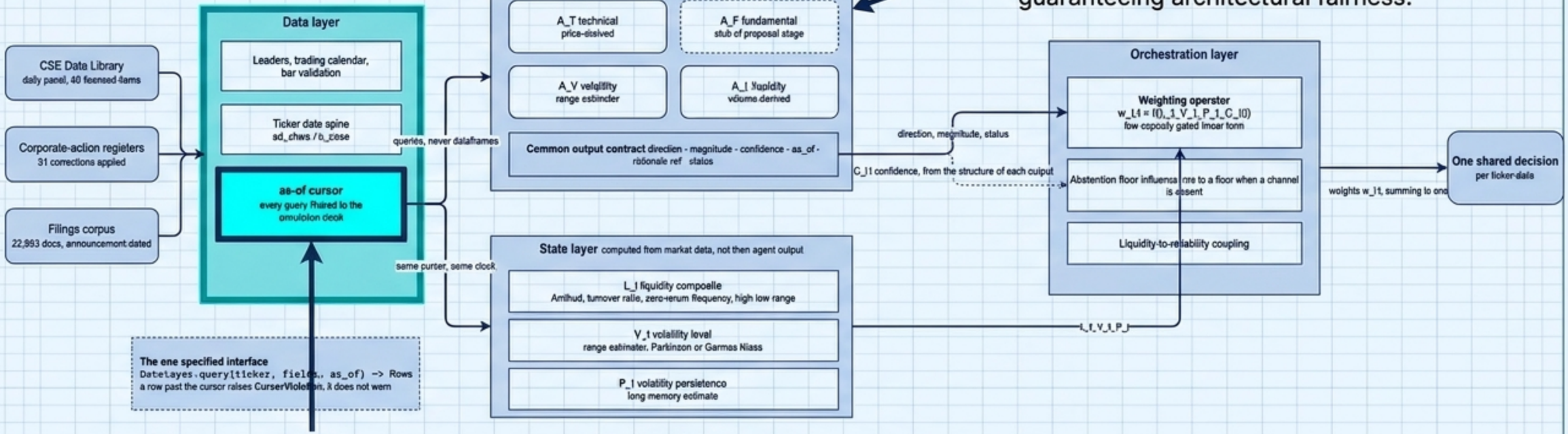
One decision for that ticker, that day

Abstention Floor (phi): (1 - K*phi)*w_{i,t} + phi

Engineering Decision: Influence falls to a floor rather than zero when a channel is absent. Unparseable output drops to a hard zero.

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System Architecture & The As-Of Cursor



Cached Execution: Agents run once; outputs are cached so 5 evaluation **arms** consume byte-identical data, guaranteeing architectural fairness.

The Non-Negotiable Rule: Every component queries a single clock. A row past the cursor raises CursorViolation—it does **not** warn. This guarantees point-in-time correctness and prevents backtest leakage seen in prior CSE literature.

Reading the diagram
An agent path: what each agent believes. A state path: what the market is doing.
The two paths share a data source and some share an output path, so no agent's own view feeds the weight it receives.
Dashed border marks a component not yet built. Dashed arrow marks a side channel.

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Agent Heterogeneity & Structural Confidence

A_T (Technical): Price-derived, gated by Theil's U2 >= 1.

A_V (Volatility): Range estimators, confidence via interval width.

A_F (Fundamental): 22,893 filings, confidence via step-count difficulty.

A_L (Liquidity): Volume-derived, confidence via rolling loss ratio.

Common Output Contract

```
...
struct CommonOutputContract {
    ticker: string; // e.g., "AAPL"
    date: date; // YYYY-MM-DD
    raw_score: float; // agent-specific
    calibrated_confidence: float; // 0.0 to 1.0, via empirical accuracy-at-threshold
    direction: int; // -1 (Sell), 0 (Hold), 1 (Buy)
    magnitude: float; // normalized expected return
    status: enum; // ACTIVE, ABSTAIN, ERROR
    metadata: map<string, any>; // agent-specific diagnostics
}
```

The Engineering Problem & Solution

A softmax and a posterior variance are not comparable.
Solution: Every agent maps its raw score through an empirical accuracy-at-threshold curve on a rolling window to share a single, commensurable unit in the output contract.

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Requirements Traceability

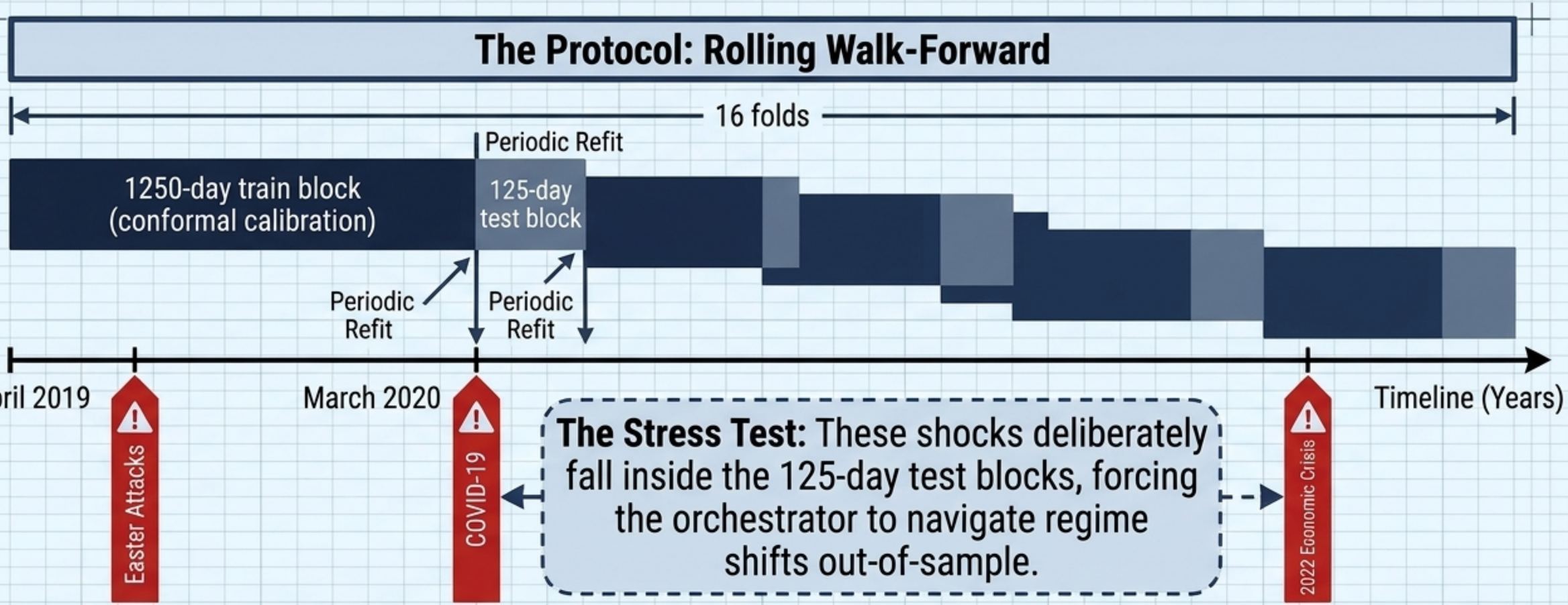
The Traceability Chain

Goal	Requirement	Evidence
Prevent lookahead bias in historical simulation.	**FR1/NFR1:** Single as-of cursor at the data layer.	Automated CursorViolation exception log.
Guarantee differences are due strictly to weighting.	**NFR2:** Cached outputs; agents run once.	Byte-identical agent outputs across all 5 arms.
Explain specific decisions post-hoc (Contribution K7).	**NFR7:** Ticker-date logging of states and weights.	Random decision perfectly reconstructed from logs alone.

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Evaluation Harness: The 16-Fold Protocol

The Protocol: Rolling Walk-Forward

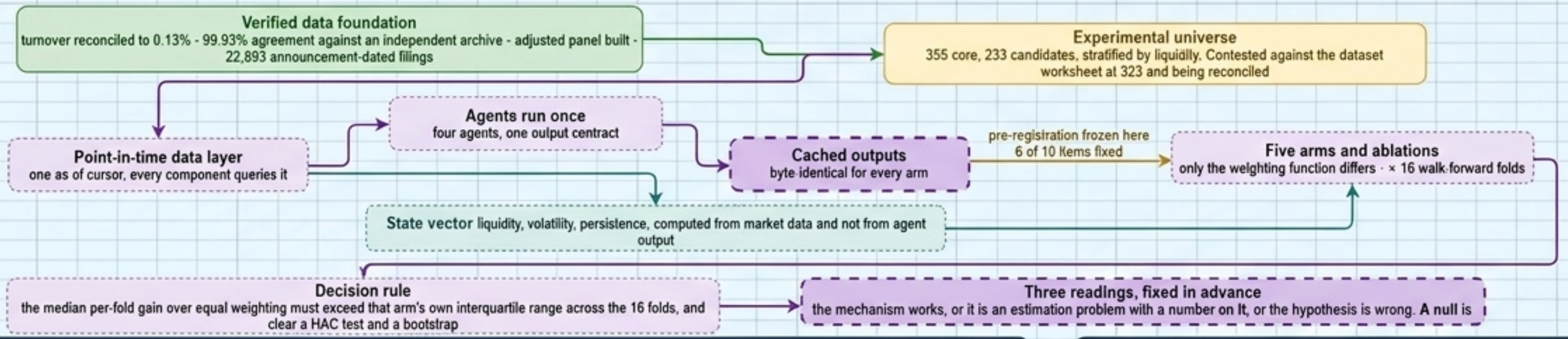


The Stress Test: These shocks deliberately fall inside the 125-day test blocks, forcing the orchestrator to navigate regime shifts out-of-sample.

Metric Constraint: Abstention is evaluated economically (abstained days held flat, returning cost-adjusted metrics alongside class balance and MCC).

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Experimental Baselines & The Oracle



The Arms

- **B_1**: Equal Weight (The pre-registered primary benchmark)
- **B_2**: Static Learned Weight
- **B_3**: Discounted Inverse-MSE (The real threat: Tests if recent performance alone is enough without market state)
- **P:** The Proposal (State-conditioned)
- **O:** Oracle (Fitted on test window)

The Oracle Insight: O is not a strategy. It is a hindsight diagnostic bound to separate a wrong hypothesis from a hard estimation problem.

The Decision Rule

Median per-fold Delta_k > IQR(B_1), plus HAC-adjusted paired test and stationary bootstrap at alpha = 0.05.

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Data Strategy: The Colombo Stock Exchange

Why CSE?
It's a frontier-market stress-test where reliability varies wildly (thin channels, structural breaks).

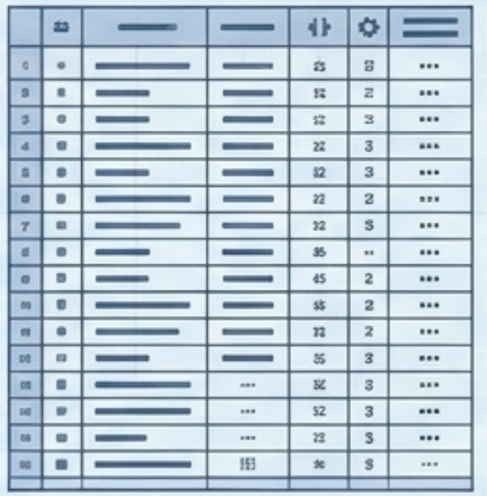
Raw Registers



Hygiene & Corrections

- ✓ 31 corporate-action register corrections applied.
- ✓ 259 events marked as approximate in the registers, dropping/flagging them for daily-return models.
- ✓ Total-return (tr_close) used as primary against ASTRI.
- ✓ Reconciled against independent archives at 99.93% close agreement across 817,000 rows.

Adjusted Point-in-Time Panel

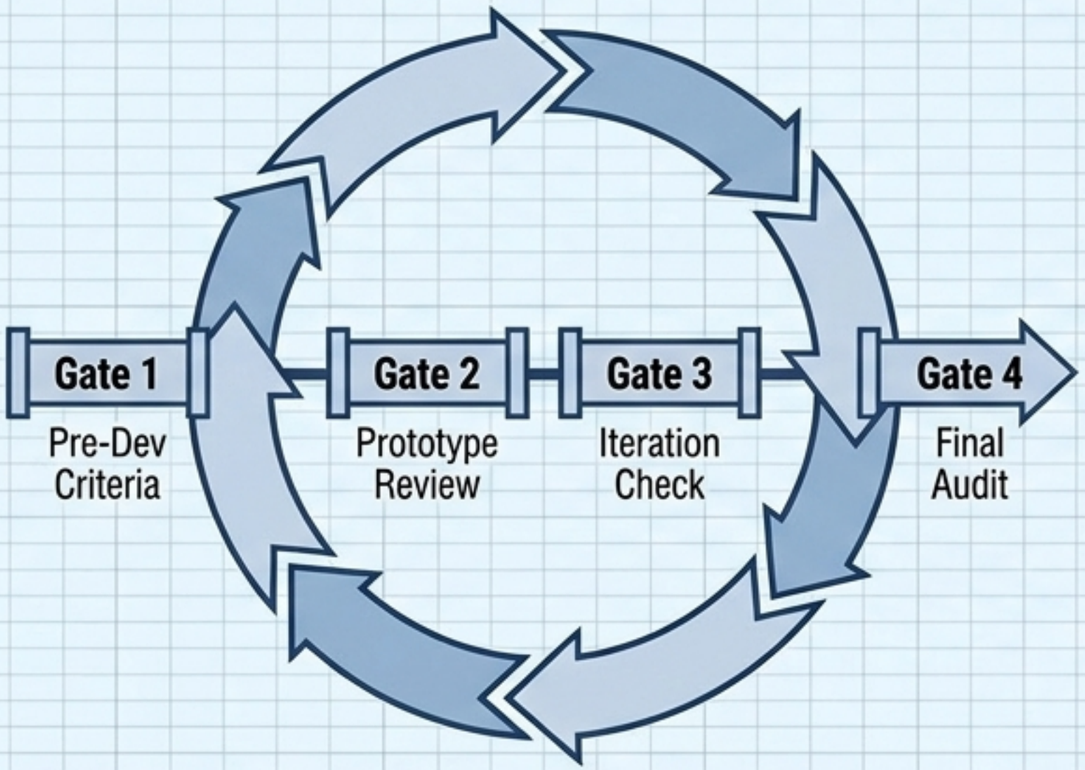


Survivorship Proxy: No listing register with effective dates exists; series-end dates stand in as a proxy, and residual bias is explicitly reported.

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Development Methodology & Risk Management

Methodology



Evolutionary prototyping managed through strict stage gates. Acceptance criteria are written before each increment.

AI Verification Audit

Lead with verification, not assistance.

1,621

quoted spans traced perfectly to original papers.

13

substantive errors caught in paper reviews (4 adjudicated directly against source PDFs).

"A verification process that has never caught anything is not evidence of correctness. The audit is the point."

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Project Scope & Deliberate Boundaries

In Scope	Explicitly Excluded
	- No sentiment analysis (ruled out by evidence). - No fifth agent (e.g., Macro). - No reinforcement learning or network orchestrators. - No end-user UI (this is a batch-run research pipeline). - No unvalidated EDGE spread estimates.

Limitation Concession: This is simulated execution using proxies for illiquidity, not a deployed recommender system. Furthermore, a 1250-day train window removes the thinnest counters from conformal calibration.

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Closing Defense

ADDRESSES:

How the influence of methodologically heterogeneous agents should adapt to varying evidence reliability.

NEEDED BECAUSE:

Existing systems fix influence at design time, failing silently with high confidence when channels go thin.

PROPOSED CONTRIBUTION:

A continuous, exogenous-state-conditioned weighting operator with an abstention channel.

CREDIBLE BECAUSE:

Tested via a punishing 16-fold walk-forward evaluation protocol where equal-weighting is the pre-registered, acceptable-null primary baseline.

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